

LEVISAT 500 MG/5 ML Concentrate for solution for infusion

DESCRIPTION: Levistat is a clear colorless sterile solution free from visible particulate matter.

Before administration, Levistat concentrate for solution for infusion can be diluted with the following diluents;

- 5% Dextrose Injection
- 0.9% Sodium chloride injection

Description after dilution: After dilution, the solution remains clear and colourless without any visible foreign matter.

Note: No diluent is provided along with Levistat concentrate for solution for infusion.

COMPOSITION:
Each 5 ml vial contains Levistatracetam 500mg.

PHARMACODYNAMICS:
Pharmacotherapeutic group: Antiepileptic,
Mechanism of action

The active substance, levistatracetam, is a pyrrolidone derivative (S-enantiomer of α -ethyl-2-oxo-1-pyrrolidone acetamide), chemically unrelated to existing antiepileptic active substances. The mechanism of action of levistatracetam still remains to be fully elucidated.

Pharmacodynamics effects

In man, an activity in both partial and generalized epilepsy conditions (epileptiform discharge/photo paroxysmal response) has confirmed the broad spectrum pharmacological profile of levistatracetam.

PHARMACOKINETICS:

Levistatracetam is a highly soluble and permeable compound. The pharmacokinetic profile is linear with low intra- and inter-subject variability. There is no modification of the clearance after repeated administration.

Distribution

Neither levistatracetam nor its primary metabolite are significantly bound to plasma proteins (<10%). The volume of distribution of levistatracetam is approximately 0.5 to 0.7 l/kg, a value close to total body water volume.

Metabolism

Levistatracetam is not extensively metabolized in humans. The major metabolic pathway (24 % of the dose) is an enzymatic hydrolysis of the acetamide group. Production of the primary metabolite, ucb L057, is not supported by liver cytochrome P₄₅₀ isoforms. Hydrolysis of the acetamide group was measurable in a large number of tissues including blood cells. The metabolite ucb L057 is pharmacologically inactive.

Elimination

The plasma half-life in adults was 7±1 hours and did not vary either with dose, route of administration or repeated administration. The mean total body clearance was 0.96ml/min/kg. It is therefore recommended to adjust the maintenance daily dose of levistatracetam, based on creatinine clearance in patients with moderate and severe renal impairment.

Renal impairment

The apparent body clearance of both levistatracetam and of its primary metabolite is correlated to the creatinine clearance.

INDICATIONS:

Levistat is indicated as monotherapy in the treatment of partial onset seizures with or without secondary generalization in patients from 16 years of age with newly diagnosed epilepsy.

Levistat is indicated as adjunctive therapy for the following:

- In the treatment of partial onset seizures with or without secondary generalization in adults and children from 4 years of age with epilepsy.
- In the treatment of myoclonic seizures in adults and adolescents from 12 years of age with Juvenile Myoclonic Epilepsy.
- In the treatment of primary generalized tonic-clonic seizures in adults and children from 12 years of age with Idiopathic Generalized Epilepsy.

Levistat Concentrate is an alternative for patients (adults and children from 4 years of age) when oral administration is not feasible.

CONTRAINDICATIONS:

Levistatracetam is contraindicated in:

- Hypersensitivity to the active substance or other pyrrolidone derivatives or to any of the excipients.

SIDE EFFECTS / ADVERSE REACTIONS:

Infections and infestations

Nasopharyngitis (frequency very common); Infection (frequency rare)

Blood and lymphatic system disorders

Thrombocytopenia, leukopenia (frequency uncommon); Pancytopenia, neutropenia, agranulocytosis (frequency rare)

Immune system disorders

Drug reaction with eosinophilia and systemic symptoms (DRESS), hypersensitivity (including angioedema and anaphylaxis) (frequency rare)

Metabolism and nutrition disorders

Anorexia (frequency common); Weight decreased, weight increase (frequency uncommon); Hyponatremia (frequency rare)

Psychiatric disorders

Depression, hostility/aggression, anxiety, insomnia, nervousness/irritability (frequency common); Suicide attempt, suicidal ideation, psychotic disorder, abnormal behavior, hallucination, anger, confusional state, panic attack, affect lability/mood swings, agitation (frequency uncommon); completed suicide, personality disorder, thinking abnormal, delirium (frequency rare)

Nervous system disorders

Somnolence, headache (frequency very common); Convulsion, balance disorder, dizziness, lethargy, tremor (frequency common); Amnesia, memory impairment, coordination abnormal/ataxia, paraesthesia, disturbance in attention (frequency uncommon); Choroathetosis, dyskinesia, hyperkinesia, gait disturbance, choreoathetosis (frequency rare)

Eye disorders

Diplopia, vision blurred (frequency uncommon)

Ear and labyrinth disorders

Vertigo (frequency common)

Respiratory, thoracic and mediastinal disorders

Cough (frequency common)

Gastrointestinal disorders

Abdominal pain, diarrhea, dyspepsia, vomiting, nausea (frequency common); Pancreatitis (frequency rare)

Hepatobiliary disorders

Liver function test abnormal (frequency uncommon); Hepatic failure, hepatitis (frequency rare)

Renal and urinary disorders

Acute kidney injury (frequency rare)

Skin and subcutaneous tissue disorders

Rash (frequency common); Alopecia, eczema, pruritis (frequency uncommon); Toxic epidermal necrolysis, Stevens-Johnson syndrome, erythema multiforme (frequency rare)

Musculoskeletal and connective tissue disorders

Muscular weakness, myalgia (frequency uncommon); Rhabdomyolysis and blood creatinine phosphokinase increased* (frequency rare)

*Prevalence is significantly higher in Japanese patients when compared to non-Japanese patients.

General disorders and administration site conditions

Asthenia/fatigue (frequency common)

Injury, poisoning and procedural complications

Injury (frequency uncommon)

Cases of encephalopathy have been rarely observed after levistatracetam administration. These undesirable effects generally occurred at the beginning of the treatment (few days to a few months) and were reversible after treatment discontinuation.

PRECAUTIONS / WARNINGS:

Discontinuation

If levistatracetam has to be discontinued it is recommended to withdraw it gradually (e.g. in adults and adolescents weighing more than 50kg: 500 mg decreases twice daily every two to four weeks; in children and adolescents less than 50kg: dose decrease should not exceed 10mg/kg twice daily every two weeks).

Renal or hepatic impairment

The administration of levistatracetam to patients with renal impairment may require dose adjustment. In patients with severely impaired hepatic function, assessment of renal function is recommended before dose selection.

Acute Kidney injury

The use of levistatracetam has been very rarely associated with acute kidney injury, with a time to onset ranging from a few days to several months.

Blood cell counts

Rare cases of decreased blood cell counts (neutropenia, agranulocytosis, leucopenia, thrombocytopenia and pancytopenia) have been described in association with levistatracetam administration, generally at the beginning of the treatment. Complete blood cell counts are advised in patients experiencing important weakness, pyrexia, recurrent infections or coagulation disorders.

Suicide

Potential for an increase in risk of suicidal thoughts or behaviours.

Suicide, suicide attempt, suicidal ideation and behaviour have been reported in patient treated with anti-epileptic agents (including levistatracetam). A meta-analysis of randomized placebo-controlled trials of anti-epileptic medicinal products has shown a small increased risk of suicidal thoughts and behaviour. The mechanism of this risk is not known.

Therefore, patients should be monitored for signs of depression and/or suicidal ideation and behaviour and appropriate treatment should be considered. Patients (and caregivers of patients) should be advised to seek medical advice should sign of depression and/or suicidal ideation or behaviour emerge.

Abnormal and aggressive behaviours

Levistatracetam may cause psychotic symptoms and behavioural abnormalities including irritability and aggressiveness. Patients treated with levistatracetam should be monitored for developing psychiatric signs suggesting important mood and/or personality changes. If such behaviours are noticed, treatment adaptation or gradual discontinuation should be considered.

Pediatric population

Long term effects on learning, intelligence, growth, endocrine function, puberty and childbearing potential in children remain unknown.

Excipients

This medicinal product contains 2.5 mmol (or 57 mg) sodium per maximum single dose (0.8 mmol (or 19 mg) per vial). To be taken into consideration by patients on a controlled sodium diet.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINE:

Levistatracetam has minor or moderate influence on the ability to drive and use machines. Due to possible individual sensitivity, some patients might experience somnolence or other central nervous system related symptoms, especially at the beginning of treatment or following a dose increase. Therefore, caution is recommended in those patients when performing skilled tasks, e.g. driving vehicles or operating machinery. Patients are advised not to drive or use machines until it is established that their ability to perform such activities is not affected.

DRUG INTERACTIONS:

Antiepileptic medicinal products

Levistatracetam did not influence the serum concentrations of existing antiepileptic medicinal products (phenytoin, carbamazepine, valproic acid, phenobarbital, lamotrigine, gabapentin and primidone) and that these antiepileptic medicinal products did not influence the pharmacokinetics of Levistatracetam.

Probenecid

Probenecid (500 mg four times daily), a renal tubular secretion blocking agent, has been shown to inhibit the renal clearance of the primary metabolite, but not of levistatracetam. Nevertheless, the concentration of this metabolite remains low.

Methotrexate

Concomitant administration of Levistatracetam and methotrexate has been reported to decrease methotrexate clearance, resulting in increased/prolonged blood methotrexate concentration to potentially toxic levels. Blood methotrexate and Levistatracetam levels should be carefully monitored in patients treated concomitantly with the two drugs.

Oral contraceptives and other pharmacokinetics interactions

Levistatracetam 1,000 mg daily did not influence the pharmacokinetics of oral contraceptives (ethinyl-estradiol and levonorgestrel); endocrine parameters (luteinizing hormone and progesterone) were not modified. Levistatracetam 2,000 mg daily did not influence the pharmacokinetics of digoxin and warfarin; prothrombin times were not modified. Co-administration with digoxin, oral contraceptives and warfarin did not influence the pharmacokinetics of levistatracetam.

Alcohol

No data on the interaction of levistatracetam with alcohol are available.

Incompatibilities:

This medicinal product must not be mixed with other medicinal products.

INSTRUCTIONS FOR USE:

Levistatracetam, 500 mg/5ml concentrate for solution for infusion Table presents the recommended preparation and administration of levistatracetam concentrate to achieve a total daily dose of 500 mg, 1,000 mg, 2,000 mg, or 3,000 mg in two divided doses.

Dose	Withdrawn Volume	Volume of Diluent	Infusion time	Frequency of administration	Total Daily Dose
250 mg	2.5 ml (half 5 ml vial)	100 ml	15 minutes	Twice daily	500 mg/day
500 mg	5 ml (one 5 ml vial)	100 ml	15 minutes	Twice daily	1000 mg/day
1000 mg	10 ml (two 5 ml vials)	100 ml	15 minutes	Twice daily	2000 mg/day
1500 mg	15 ml (three 5 ml vials)	100 ml	15 minutes	Twice daily	3000 mg/day

This medicinal product is for single use only; any unused solution should be discarded.

This medicinal product must not be mixed with other medicinal products except those mentioned below. Levistatracetam concentrate was found to be physically compatible and chemically stable when mixed with the following diluents for at least 24 hours and stored at controlled room temperature 30°C.

Diluents:

- Sodium chloride (0.9%) injection.
- Dextrose 5% injection

Medicinal product with particulate matter or discoloration should not be used. Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

PREGNANCY AND LACTATION:

Women of child bearing potential

Advice should be given to women who are of childbearing potential. Treatment with levistatracetam should be reviewed when a woman is planning to become pregnant. As with all antiepileptic medicines, sudden discontinuation of levistatracetam should be avoided as this may lead to breakthrough seizures that could have serious consequences for the woman and the unborn child. Monotherapy should be preferred whenever possible because therapy with multiple antiepileptic medicines AEDs could be associated with a higher risk of congenital malformations than monotherapy, depending on the associated antiepileptics.

Pregnancy

Levistatracetam can be used during pregnancy, if after careful assessment it is considered clinically needed. In such case, the lowest effective dose is recommended.

Physiological changes during pregnancy may affect levistatracetam concentration. Decrease in levistatracetam plasma concentrations has been observed during pregnancy. Appropriate clinical management of pregnant women treated with levistatracetam should be ensured.



Breastfeeding

Levetiracetam is excreted in human breast milk. Therefore, breast-feeding is not recommended.

However, if levetiracetam treatment is needed during breastfeeding, the benefit/risk of the treatment should be weighed considering the importance of breastfeeding.

Fertility

No impact on fertility was detected. No clinical data are available, potential risk for human is unknown.

RECOMMENDED DOSAGE, DOSAGE SCHEDULE AND ROUTE OF ADMINISTRATION:

Posology:

Levetiracetam therapy can be initiated with either intravenous or oral administration. Conversion to or from oral to intravenous administration can be done directly without titration. The total daily dose and frequency of administration must be maintained. Levetiracetam concentrate is for intravenous use only and the recommended dose must be diluted in at least 100ml of compatible diluent and administered intravenously as a 15-minute intravenous infusion.

Levetiracetam concentrate is an alternative for patients (adults and children from 4 years of age) when oral administration is temporarily not feasible.

Adults

Monotherapy for adults and adolescents from 16 years of age

The recommended starting dose is 250 mg twice daily which should be increased to an initial therapeutic dose of 500 mg twice daily after two weeks. The dose can be further increased by 250 mg twice daily every two weeks depending upon the clinical response. The maximum dose is 1500 mg twice daily.

Add-on therapy for Adults (≥18 years) and adolescents (12 to 17 years) weighing 50 kg or more

The initial therapeutic dose is 500 mg twice daily. This dose can be started on the first day of treatment.

Depending upon the clinical response and tolerability, the daily dose can be increased up to 1500 mg twice daily. Dose changes can be made in 500 mg twice daily increases or decreases every two to four weeks.

Children

The physician should prescribe the most appropriate pharmaceutical form, presentation and strength according to age, weight and dose.

Monotherapy

The safety and efficacy of levetiracetam in children and adolescents below 16 years as monotherapy treatment have not been established. There are no data available.

Add-on therapy for children (4 to 11 years) and adolescents (12 to 17 years) weighing less than 50 kg

The initial therapeutic dose is 10 mg/kg twice daily. Depending upon the clinical response and tolerability, the dose can be increased up to 30 mg/kg twice daily. Dose changes should not exceed increases or decreases of 10 mg/kg twice daily every two weeks.

The lowest effective dose should be used.

Dose in children 50 kg or greater is the same as in adults.

Dose recommendations for children and adolescents:

Weight	Starting dose 10 mg/kg twice daily	Maximum dose 30 mg/kg twice daily
10 kg ⁽¹⁾	100 mg (1 ml) twice daily	300 mg (3 ml) twice daily
15 kg ⁽¹⁾	150 mg (1.5 ml) twice daily	450 mg (4.5 ml) twice daily
20 kg ⁽¹⁾	200 mg (2 ml) twice daily	600 mg (6 ml) twice daily
25 kg	250 mg twice daily	750 mg twice daily
From 50 kg ⁽²⁾	500 mg twice daily	1500 mg twice daily

(1) Children 25 kg or less should preferably start the treatment with levetiracetam 100 mg/ml.

(2) Dose in children and adolescents 50 kg or more is the same as in adults

Special populations

Elderly (65 years and older)

Adjustment of the dose is recommended in elderly patients with compromised renal function.

Renal impairment

The daily dose must be individualized according to renal function. For adult patients, refer to the following table and adjust the dose as indicated. To use this dosing table, an estimate of the patient's creatinine clearance (CLcr) in mL/min is needed. The CLcr in mL/min may be estimated from serum creatinine (mg/dL) determination, for adults and adolescents weighing 50 kg or more, the following formula:

$$CLcr (mL/min) = \frac{[140 - \text{age (years)}] \times \text{weight kg}}{72 \times \text{serum creatinine (mg/dl)}} \quad (\times 0.85 \text{ for women})$$

Then CLcr is adjusted for body surface area (BSA) as follows:

$$CLcr (mL/min/1.73m^2) = \frac{CLcr (mL/min)}{BSA \text{ subject (m}^2\text{)}} \times 1.73$$

Dosing adjustment for adult and adolescent's patients weighing more than 50 kg with impaired renal function

Group	Creatinine clearance (mL/min/1.73 m ²)	Dose and frequency
Normal	≥ 80	500 to 1,500 mg twice daily
Mild	50-79	500 to 1,000 mg twice daily
Moderate	30-49	250 to 750 mg twice daily
Severe	< 30	250 to 500 mg twice daily
End-stage renal disease patients undergoing dialysis ⁽¹⁾	--	500 to 1,000 mg once daily ⁽²⁾

(1) A 750 mg loading dose is recommended on the first day of treatment with levetiracetam

(2) Following dialysis, a 250 to 500 mg supplemental dose is recommended.

For children with renal impairment, levetiracetam dose needs to be adjusted based on the renal function as levetiracetam clearance is related to renal function. This recommendation is based on a study in adult renally impaired patients.

The CLcr in mL/min/1.73 m² may be estimated from serum creatinine (mg/dL) determination using, for young adolescents and children using the following formula (Schwartz formula):

$$CLcr (mL/min/1.73m^2) = \frac{\text{Height (cm)} \times \text{ks}}{\text{Serum creatinine (mg/dl)}}$$

ks= 0.55 in Children to less than 13 years and in adolescent female; ks= 0.7 in adolescent male

Dosing adjustment for children and adolescents patients weighing less than 50 kg with impaired renal function

Group (mL/min/1.73 m ²)	Creatinine clearance Dose and frequency	Dose and frequency
		Children from 4 years and adolescents weighing less than 50 kg
Normal	≥ 80	10 to 30 mg/kg (0.10 to 0.30 mL/kg) twice daily
Mild	50-79	10 to 20 mg/kg (0.10 to 0.20 mL/kg) twice daily
Moderate	30-49	5 to 15 mg/kg (0.05 to 0.15 mL/kg) twice daily
Severe	< 30	5 to 10 mg/kg (0.05 to 0.10 mL/kg) twice daily

(1) A 15 mg/kg (0.15 mL/kg) loading dose is recommended on the first day of treatment with levetiracetam.

(2) Following dialysis, a 5 to 10 mg/kg (0.05 to 0.10 mL/kg) supplemental dose is recommended.

Hepatic impairment

No dose adjustment is needed in patients with mild to moderate hepatic impairment. In patients with severe hepatic impairment, the creatinine clearance may underestimate the renal insufficiency. Therefore, a 50 % reduction of the daily maintenance dose is recommended when the creatinine clearance is < 60 mL/min/1.73 m².

SYMPTOMS AND TREATMENT FOR OVERDOSE, AND ANTIDOTE(S):

Symptoms

Somnolence, agitation, aggression, depressed level of consciousness, respiratory depression and coma were observed with Levetiracetam overdose.

Management of overdose

There is no specific antidote for levetiracetam. Treatment of an overdose will be symptomatic and may include haemodialysis. The dialyzer extraction efficiency is 60 % for levetiracetam and 74 % for the primary metabolite.

PACKING / PACK SIZES:

10's x 5mL, clear USP Type I glass vial with 20mm grey butyl rubber stopper and 20mm green flip off aluminum seal/ Box.

STORAGE CONDITIONS, USER INSTRUCTIONS AND PHARMACEUTICAL PRECAUTIONS:

Store below 30°C.

SHELF LIFE:

2 years

Storage Condition After Dilution:

Please use within 24 hours, store below 30°C. Balance solution or unused solution should be discarded.

MANUFACTURER AND PRODUCT REGISTRATION HOLDER

SM PHARMACEUTICALS SDN BHD (218620-M)

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